

An adaptive dual-level reinforcement learning approach for optimal trade execution

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ABSTRACT

The purpose of this research is to devise a tactic that can closely track the daily cumulative volume-weighted average price (VWAP) using reinforcement learning while minimizing the deviation from the VWAP. Previous studies often choose a relatively short trading horizon to implement their models, making it difficult to accurately track the daily cumulative VWAP since the stock price movement is often insignificant within the short trading horizon. On the other hand, training reinforcement learning models directly over a longer, daily horizon is burdensome due to extensive sequence length. Hence, there is a need for a method that can divide the long daily horizon into smaller, more manageable segments. We propose a method that leverages the U-shaped pattern of intraday stock trade volumes and uses Proximal Policy Optimization (PPO) as the learning algorithm. Our method follows a dual-level approach: a Transformer model that captures the overall (global) distribution of daily volumes in a U-shape, and a LSTM model that handles the distribution of orders within smaller (local) time intervals. The results from our experiments suggest that this dual-level architecture improves cumulative VWAP tracking accuracy compared to previous reinforcement learning approaches. The key finding is that explicitly accounting for the U-shaped intraday volume pattern leads to better performance in approximating the cumulative daily VWAP. This has implications for developing trading strategies that need to efficiently track VWAP over a full trading day.

1. Introduction

The optimal trade execution problem aims to find a strategy to optimally trade large orders within a given period of time. One of the most common and practical methods that practitioners frequently use is known as Volume Weighted Average Price (VWAP) trading (Madhavan, 2002). Research regarding the optimality of VWAP as an execution strategy is discussed in Kato (2015), which further highlights the importance of the ability of a trading model to track VWAP throughout time consistently. VWAP is calculated by adding up the dollars traded for every transaction (price multiplied by the number of shares traded) and then dividing by the total shares traded for the day. This gives an average price that takes into account both the price and the volume of shares traded.

Funds and traders often use VWAP as a benchmark to compare the price at which they executed trades to the overall market price for the security, in order to evaluate the performance of their trades. Suppose a fund intends to significantly change its stock holdings but

wants to avoid market impact and the risk of front-runners exploiting its strategy. To achieve this objective, the fund may outsource the trading to external brokerages or trading firms. In such a scenario, VWAP is commonly the most relevant metric for quantifying trading performance. If the typical end-of-day market price were used as the performance metric, traders might execute all orders at the market's close. This could lead the fund to suffer from market impact, and the end-of-day market price might become more volatile. Since VWAP reflects a stock's transaction price over time, it serves as a more suitable metric for trading performance. Using VWAP allows the fund to execute its strategy with minimal market impact, thereby avoiding potential losses from market speculation and front-running.

In Bertsimas and Lo (1998), a fundamental work for optimal trade execution is proposed, where the authors assume that market prices follow an arithmetic random walk. They use a dynamic programming principle to find an explicit closed-form solution. Building on this work, Huberman and Stanzl (2005) and Almgren and Chriss (2001)

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extended the result in [Bertsimas and Lo \(1998\)](#) by incorporating transaction costs, more complex price impact functions, and risk aversion parameters, under the assumption that market prices follow a Brownian motion. These dynamical approaches, however, are difficult to apply directly in the real world due to the discrepancy between their strong market assumptions and the complexity of actual market conditions.

On the other hand, both practitioners and researchers have widely used the time-weighted average price (TWAP) strategy and the volume-weighted average price (VWAP) strategy ([Berkowitz et al., 1988](#); [Kakade et al., 2004](#)), which are based on either pure rules or statistical rules. Especially in [Kakade et al. \(2004\)](#), the authors used historical data to estimate the average volume traded for each time interval and split the order accordingly. However, this strategy is not well-suited for capturing unexpected volatility.

From the perspective that optimal execution problems are a type of sequential decision-making task, RL has been commonly applied in this field as it is a type of stochastic decision-making process that automates the practitioner's task of using past data to make decisions on when to execute orders. Since RL approaches have many advantages, e.g. capable of capturing the market's microstructure, the trading results conducted by RL solutions often outperform traditional VWAP tracking approaches such as [Bialkowski et al. \(2008\)](#) and [Podobnik et al. \(2009\)](#). To the best of our knowledge, [Nevmyvaka et al. \(2006\)](#) is the first work to leverage RL frameworks such as Q-learning ([Watkins & Dayan, 1992](#)) to optimal trade execution problems. It is important to note that the curse of dimensionality in Q-learning makes it challenging to handle high-dimensional data. While ([Hendricks & Wilcox, 2014](#)) attempts to combine RL with the Almgren–Chriss model, it is a challenging task as the model depends on certain assumptions about the market dynamics.

Thanks to the advancements in deep RL, recent studies such as [Macri and Lillo \(2024\)](#), [Ning et al. \(2021\)](#), and [Lin and Beling \(2019\)](#) have utilized Deep Q-Networks (DQNs) ([Mnih et al., 2013](#)) for optimal trade execution, addressing the challenges of high-dimensional data and the complexity of the financial market without relying on any market assumptions. However, these methods require the design of specific attributes, which can be labor-intensive. Recently, several studies have investigated the use of proximal policy optimization (PPO) ([Schulman et al., 2017](#)) based on optimal execution frameworks, which do not require manually designed feature engineering. For instance, [Lin and Beling \(2020\)](#), [Fang et al. \(2021\)](#), [Pan et al. \(2022\)](#), and [Byun et al. \(2023\)](#) have explored this approach in different scenarios. Specifically, [Pan et al. \(2022\)](#) focused on optimal execution with limit orders, while [Lin and Beling \(2020\)](#) and [Fang et al. \(2021\)](#) used market orders. A method combining both limit and market orders was proposed by [Byun et al. \(2023\)](#).

Prior literature often considers a relatively short trading horizon or the entire day for RL implementation. However, VWAP approximation becomes challenging when a short time frame is used since the variations of financial data are often insignificant within short time intervals. Moreover, the naive approach of considering the entire day can put a strain on the network architecture due to the long sequence length. This paper aims to develop a strategy that can consistently track the daily cumulative VWAP. To achieve this goal, we start with the observation of an important stock trading characteristic, the U-shaped intraday trading pattern, documented in [Jain and Joh \(1988\)](#) and [Goodhart and O'Hara \(1997\)](#). This is a well-known stylized fact stating that a stock's trading volume often follows a U-shaped pattern throughout the day. This means that volume is highest at the opening, falls rapidly to lower levels, and then rises again towards the close of the market, but is relatively low in the afternoon. (See [Fig. 1](#)). In view of this observation, we propose a novel dual-level approach to minimize the market impact and track the daily cumulative VWAP accurately and consistently. As the name suggests, our model consists of two stages. In the first stage, the model decides how much trade volume will be executed in each interval, using the U-shape property as a guideline. In the second stage, the RL model decides how the orders

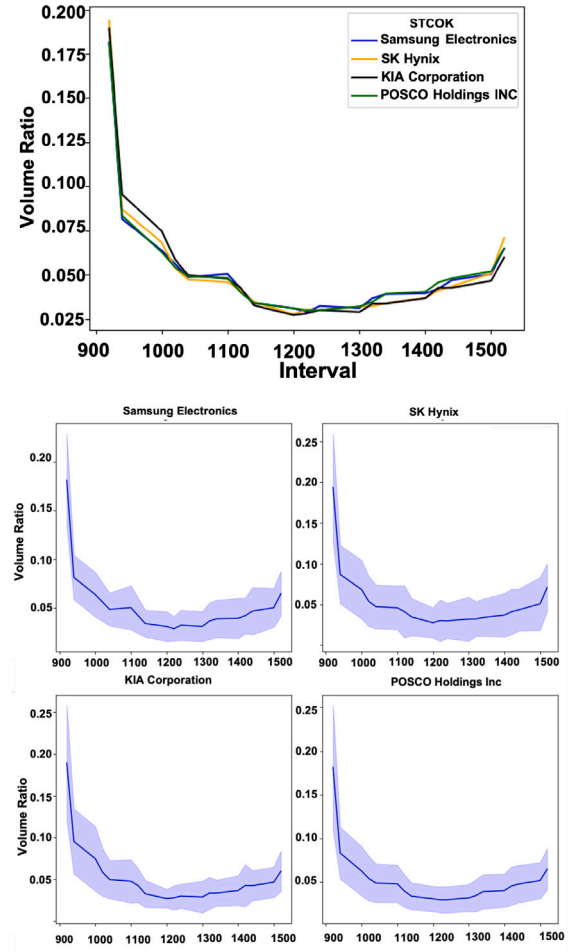


Fig. 1. Trade volume ratio over each 20-minute period throughout the day. The lines represent the 1-year averages of the ratios and the shaded regions are drawn from daily deviations from the averages.

will be executed in each interval. We propose two methods for the first stage, the statistical U-shape method and the U-shape Transformer method. The statistical U-shape method allocates the volume based on the historical average trade volumes. However, this approach cannot fully capture the day-to-day variations (See [Fig. 1](#)). Alternatively, we propose the U-shape Transformer for figuring out these variations. In the next stage, we apply LSTM, trained by the RL framework to distribute orders efficiently within each interval. It is important to note that there are numerous RL training algorithms, including Trust Region Policy Optimization (TRPO) ([Schulman et al., 2015](#)), Deep Q-Networks (DQN) ([Mnih et al., 2013](#)), and Proximal Policy Optimization (PPO) ([Schulman et al., 2017](#)). Among these options, we specifically choose PPO as our training algorithm. For a comprehensive understanding of these techniques and their applications, we refer to [Sutton and Barto \(2018\)](#).

We expect that this dual-level approach improves the accuracy of approximating the cumulative VWAP by properly distributing the total order. Our contributions are as follows:

1. We propose a novel approach for optimal trade execution that utilizes a dual-level strategy. Orders are allocated through two stages: in the first stage we utilize the U-shaped pattern of intraday volumes and in the second stage we implement deep reinforcement learning. Since this novel dual approach segments the data, we not only make reinforcement learning more computationally tractable but also improve prediction accuracy compared to applying RL directly on full intraday sequences.

Table 1

The table summarizes previous works on optimal trade execution utilizing the RL framework. Note that DQN and PPO are learning algorithms of deep RL. Notably, our work distinguishes itself by focusing on a long-term horizon.

	RL method	Horizon
Nevmyvaka et al. (2006)	Q-learning	Short
Hendricks and Wilcox (2014)	Q-learning	Short
Lin and Beling (2019)	DQN	Short
Ning et al. (2021)	DQN	Short
Lin and Beling (2020)	PPO	Short
Fang et al. (2021)	PPO	Short
Pan et al. (2022)	PPO	Short
This work	PPO	Long

2. We further propose the U-shape Transformer model to capture the day-to-day oscillations of the intraday U-shape distribution. While the U-shaped pattern is well-known, relying on fixed statistical estimates fails to adapt to sudden fluctuations or shifts in market conditions. The U-shape Transformer directly learns a robust representation from recent data, leveraging self-attention to automatically adapt to sudden dynamics in the intraday U-shape pattern.
3. We conduct ablation studies to show the strengths of our methods in approximating the daily cumulative VWAP. Furthermore, we demonstrate that the methods employed to distribute volumes from long trading horizons into short horizons can have a significant impact. In addition, we introduce the VAA, which provides a more practical way to evaluate how well a trading strategy tracks the daily VWAP. Our proposed dual-level approach combining U-shape Transformers and RL achieves state-of-the-art performance on the VAA (see Table 1).

2. Backgrounds

In this section, we discuss the basics of the limit order book (LOB) and the optimal trade execution process. Most modern financial exchanges, including the Korea Stock Exchange (KRX), offer electronic trading platforms with access to limit order books. Traders often rely on the information provided by the limit order book to develop successful trading strategies.

2.1. Limit order book

In financial markets, a limit order is an order to buy or sell a fixed number of shares at a specified price. This order becomes part of the limit order book, which records all limit orders for a specific stock. The bid price is the highest price a buyer is willing to pay for the security, and the ask price is the lowest price in which the seller is willing to sell. A market order, by contrast, is an order to buy or sell a fixed number of shares at the current market price without stating the price. When a market order is placed, it is matched by the best available limit orders in the LOB. The LOB is important for optimal trade execution and helps traders track supply and demand for a stock, making it easier to identify the best time and price to buy or sell shares.

2.2. Optimal trade execution

Optimal trade execution refers to the process of buying or selling a financial asset while achieving the desired objective. Generally, this problem formulates as follows: Within a timeframe of T timesteps, $0, 1, \dots, T-1$, a trader who possesses an inventory of O shares must buy or sell the entirety of the inventory. For each timestep $t \in \{0, 1, \dots, T-1\}$, the trader determines the number of shares to order O_t based on the information from LOB, and carries out the trade at an execution price p_t . For the group of traders seeking to maximize trading profit, their

objective is to find a strategy that maximizes or minimizes the average execution price $\bar{P}$, which is calculated as

$$\bar{P} = \sum_{t=0}^{T-1} \frac{O_t}{O} p_t.$$

While many financial firms pursue profits through trading, there are also numerous firms within the financial industry that prioritize consistently tracking the VWAP over maximizing trading profit. The focus of our paper is tailored for this second group of traders. The VWAP is calculated as

$$\text{VWAP} = \frac{\sum_{t=0}^{T-1} p_t q_t}{\sum_{t=0}^{T-1} q_t},$$

where q_t is the execution volume determined by the market. Therefore, the primary objective of this paper is to minimize the difference between the VWAP and $\bar{P}$, using a dual-level approach.

3. Method description

In this section, we describe our dual-level approach and formulation of the optimal trade execution problem under the lens of reinforcement learning. We also provide details on the architecture of the Transformer and LSTM neural networks and how they are integrated into our proposed method.

3.1. Dual-level approach

Our approach involves two stages of distributing orders. In the first stage, we allocate the total daily orders O into L intervals. In the second stage, we further divide the orders from each interval into smaller executable orders using RL. Our approach begins by dividing the market hours of a day into L intervals. For each interval l , we let O^l as the total order to be executed, ensuring that the sum of all interval orders equals the total daily order, or $\sum_{l=0}^{L-1} O^l = O$. We suggest two methods for implementing the first stage. The naive approach is to adhere to the statistical U-shape trade volume distribution for each interval obtained from past data (e.g. the historical average of the past year). A more advanced method would be to use the output (predicted U-shape distribution) generated by a U-shape Transformer model. In the latter case, O^l is calculated progressively by referring only to market information from past days and previous intervals within the day. For such an l th interval, which is comprised of T timesteps, the LSTM model outputs O_t^l , which is the number of orders to be executed at timestep t such that $\sum_{t=0}^{T-1} O_t^l = O^l$. Once O_t^l is decided, it is averaged over I execution steps in the corresponding i th subinterval. In other words, for each execution step, which is set as every 5 s, the final number of orders executed is O_t^l/I . The choice of averaging is supported by our findings in Section 4.2 that there is no significant impact in approximating the VWAP since the final interval is too small. Furthermore, in Section 4.2, we verify that incorporating long-term horizons yields superior efficiency. Therefore, although a deep reinforcement learning framework can offer marginal performance improvements, we have opted for the simplicity of using the uniform distribution for the final interval. A schematic illustration of this design is provided in Fig. 2.

3.2. MDP formulation for optimal execution

In this section, we explain our Markov Decision Process (MDP) formulation of optimal trade execution. A MDP is typically represented using the tuple $(S, \mathcal{A}, P, r, \gamma)$, where S is the state space, $\mathcal{A}$ is the action space, P is the transition probability, r is the reward function, and γ is the discount factor. Our configurations of the state space S and action space $\mathcal{A}$ are similar to that of Lin and Beling (2020). It is worth noting that the MDP implementation in this paper is applied on a per l th interval basis for a given day.

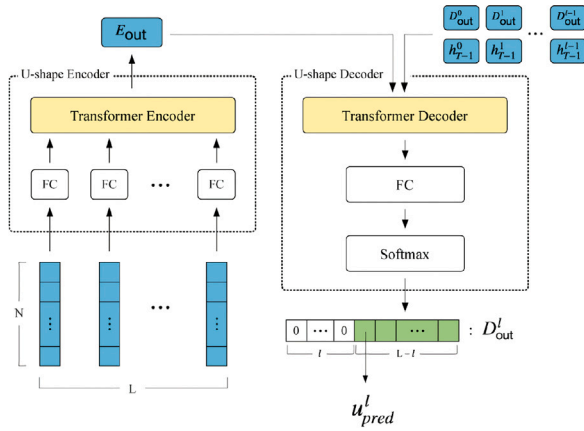


Fig. 3. The architecture of the U-shape Transformer. The U-shape Decoder in this figure represents the l th decoding step.

day-to-day variations of the distribution. The U-shape Decoder is composed of a Transformer Decoder and L linear layers, each mapping to different dimensions. In contrast to $\mathcal{E}$, these linear layers are applied to the output of the Transformer Decoder.

For every l th ($l > 0$) decoding step, both E_{out} and the cumulative input sequence are fed as input to the Transformer Decoder, D . We denote the cumulative input sequence as $D_{in}^l = (D_{in,j}^l)_{0 \leq j \leq l-1}$, where $D_{in,j}^l := \text{CONCAT}(D_{out}^j, h_{T-1}^j)$. Here, D_{out}^j represents the output of D at the j th step, and h_{T-1}^j is the last LSTM hidden vector from the j th interval. The output from the Transformer Decoder at the l th decoding step is a vector of dimension $L + H$, which is then passed through a linear layer that reduces its dimension to $L - l$. The softmax function is applied to predict the ratios of the remaining $L - l$ intervals, and the first l components are zero-padded. Note that we use h_{T-1}^l to capture the day-to-day variations in the U-shape distribution, as it holds interval-specific information for that particular day. To ensure $\sum_{l=0}^{L-1} u_{pred}^l = 1$, the final l th interval volume ratio prediction is calculated as

$$u_{pred}^l = \left(1 - \sum_{j=0}^{l-1} u_{pred}^j\right) \cdot D_{out}^l[l]. \quad (4)$$

It is worth noting that D_{in}^0 is constructed by applying an additional linear layer that transforms the dimension of the vector holding the top 5 bid/ask volume averages of pre-market data to $L + H$. Therefore, with the given day total order denoted as O , the corresponding interval order O^l can be determined as the product of O and u_{pred}^l , i.e.

$$O^l = O \times u_{pred}^l. \quad (5)$$

3.3.2. Level 2: LSTM for local order distribution

The objective of the LSTM model is to optimally allocate orders within the intervals of a given day. The structure of our LSTM model, which is outlined in Fig. 4, is similar to that of Lin and Beling (2020). Given s_t^l as input, it first passes through two linear layers with hidden dimension 128 and is then concatenated with a_{t-1}^l and $r_{t-1}^l := r(a_{t-1}^l)$. The resulting vector is then processed by an LSTM Cell with hidden dimension H to obtain h_t^l , which is then passed separately through two linear layers to produce the policy $\pi(\cdot|s_t^l)$ and the value $V(s_t^l)$ respectively. We note that the vector h_{T-1}^l which is obtained at the last timestep is used as input to the Transformer Decoder. The action a_t^l , reward r_t^l , and the number of orders to execute O_t^l is calculated as is described in Section 3.2. The LSTM Cell also receives h_{t-1}^l and c_{t-1}^l as input and its output includes c_t^l as well.

3.3.3. Training procedure

U-shape Transformer. Since the goal of the first level of order allocation is to approximate the U-shape distribution while also tracking the daily cumulative VWAP, the training objective of the Transformer model is to minimize

$$J_{TF} := \mathbb{E}_{days} \left[\frac{c_1}{L} \sum_{l=0}^{L-1} (u_{true}^l - u_{pred}^l)^2 + c_2 VAA \right] \quad (6)$$

where c_1 and c_2 are coefficients and

$$VAA := \left| \frac{MP_{day} - VWAP_{day}}{VWAP_{day}} \right| \quad (7)$$

denotes the VWAP Approximation Accuracy (VAA), which we will further use as a general metric for performance evaluation in experiments, and

$$MP_{day} := \sum_{l=0}^{L-1} \sum_{t=0}^{T-1} \frac{O_t^l}{O} p_t^l$$

is the price yielded by our model. This is one of the measures used in practice to gauge how close a trader has traded closer to their benchmark, the VWAP. If a trader has traded at a price that is exactly equal to the VWAP, then $VAA = 0$. Thus lower VAA would be better. The first term inside the expectation of (6) learns to make ratio predictions closer to ground-truth values. The second term penalizes the Transformer model if the final daily acquisition price generated by the combination of the Transformer and LSTM models deviates from the daily VWAP. The primary objective of the U-shaped transformer in the first level is to distribute the daily total order with precision. However, relying solely on the first term of the loss function may not be enough to accomplish this objective. This is because u_{true}^l does not consider the local information, such as the LOB, which can pose difficulties in tracking the daily VWAP accurately. Furthermore, if only the first term in (6) is taken into account, the reinforcement learning framework would not influence the first level, leading to independent operation of the first and second levels. To ensure that the micro information has an appropriate impact on the first level, we incorporate the second term into the loss function.

LSTM. To find the optimal parameter θ for the LSTM model, we use an actor-critic style PPO algorithm (Schulman et al., 2017), which is one of the most popular on-policy RL algorithms, as our base learner. One of the major advantages of the PPO in this task is its ability to adapt to changing market conditions and effectively learn from noisy and high-dimensional data. This algorithm uses the actor loss $J_{PPO}^{CLIP}(\theta)$ and the critic loss function $J_{PPO}^{VF}(\theta)$, which are defined as follows:

$$J_{PPO}^{CLIP}(\theta) := \mathbb{E}_t \left[\min(q_t(\theta) \hat{A}_t, \text{clip}(q_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t) \right], \quad (8)$$

and

$$J_{PPO}^{VF}(\theta) = \mathbb{E}_t \left[(V_t^{\text{targ}} - V_\theta(s_t))^2 \right], \quad (9)$$

$$\text{where } q_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)},$$

$$V_t^{\text{targ}} = \sum_{k=t}^{T-1} \gamma^{k-t} r_k \text{ and } \hat{A}_t = V_t^{\text{targ}} - V_\theta(s_t). \quad (10)$$

PPO's goal is to find a parameter θ that maximizes the main objective function $J_{PPO}(\theta)$, which is defined by

$$J_{PPO}(\theta) = J_{PPO}^{CLIP}(\theta) - c_3 J_{PPO}^{VF}(\theta) + c_4 \mathbb{E}_t[S[\pi_\theta](s_t)], \quad (11)$$

where c_3, c_4 are coefficients, and $\mathbb{E}_t[S[\pi_\theta](s_t)]$ denotes an entropy loss term, where $S[\pi_\theta](s_t)$ is defined by

$$S[\pi_\theta](s_t) := - \sum_{a \in \mathcal{A}} \pi_\theta(a|s_t) \log \pi_\theta(a|s_t). \quad (12)$$

By incorporating an entropy term, PPO can incentivize the agent to explore alternative actions, prevent getting trapped in suboptimal policies, and mitigate overfitting to the training data (Mnih et al., 2016;

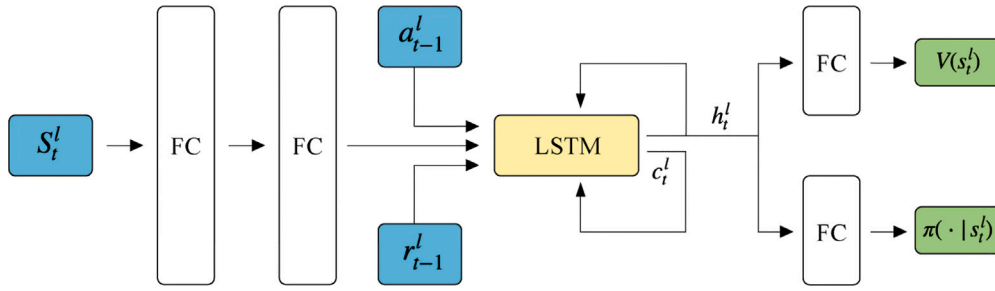


Fig. 4. The architecture of the LSTM model.

Williams, 1992). To find the parameter θ , PPO iteratively gathers episodes and updates the parameter θ using the following scheme:

$$\theta_{k+1} = \arg \max_{\theta} \mathbb{E}_s \mathbb{E}_{a \sim \pi_{\theta_k}(\cdot|s)} [J_{\text{PPO}}(\theta)], \quad (13)$$

where k stands for the k th step. The “clip” operator in (8) allows PPO to learn from previous experiences without becoming overly dependent on them. Moreover, PPO can effectively retain knowledge from past experiences while also acquiring insights from new experiences. This key attribute of PPO is one of its significant advantages.

4. Experiments

Our dual-level approach significantly enhances the accuracy and consistency of tracking the daily cumulative VWAP, despite the challenges in improving the performance of RL frameworks for optimal trade execution in relatively short trading horizons. Furthermore, the U-shape Transformer proves efficient in capturing the daily variations of the U-shape distribution. In the following section, we conduct a series of experiments to verify these assertions.

4.1. Experiment settings

4.1.1. Implementation

In all of our experiments, the hyperparameters for the loss functions, the length of the intervals, the number of total orders, and the trading horizon are fixed and set according to the following specifications:

- The coefficients for J_{TF} : $c_1 = 0.5, c_2 = 0.5$
- The coefficients for J_{PPO} : $c_3 = 1, c_4 = 0.01$
- The length of the intervals: $L = 19, T = 20$
- The number of total orders: $O \sim \mathcal{N}(2.5 \times 10^{-3} \mu, 6.25 \times 10^{-6} \sigma^2)$, where μ and σ is the average and standard deviation of the day total volume for the previous sixty days, respectively.
- Trading horizon of a day: 380 minutes (From when the market opens at 09:00:00 to when it closes at 15:20:00).

Since a typical day allows for 380 min of trading, our model first takes the daily total order, O , and breaks it down to O^l , 19 intervals of 20 min each. ($l \in [0, 18]$) We either apply the statistical U-shape method or the Transformer model for this first stage allocation process. Then, within each interval, the LSTM model further subdivides O^l into 20 sets of 1-minute subintervals, O_t^l . ($t \in [0, 19]$)

In each of these subintervals, O_t^l is divided into equal portions and executed over a span of 12 steps, assuming that orders are executed every 5 s. The method and hyperparameters used to train our agents for the experiment are summarized in Algorithms 1, 2, and Table 3 which is located at the end of the paper.

To conduct realistic simulations, we determine O in a way that takes into account the volatility of daily volume, which differs from most previous works where it is fixed to a certain value. This choice aims to simulate the fluctuations of total orders that financial firms have to execute in a day, taking into consideration recent trade volume statistics for each stock. By considering the volume statistics for each

Algorithm 1 Dual-level Neural Network

Require: Number of outer iterations and num days

Randomly initialize learnable parameters $\theta^{\mathcal{E}}$, θ^D , and $\theta^{\mathcal{L}}$

Initialize trajectory buffer $\mathcal{T}$

for $j = 1$ to outer iterations **do**

for $d = 1$ to num days **do**

 Randomly choose a date in the training set

$\mu \leftarrow$ the average of the daily volume for the last 60 days

$\sigma \leftarrow$ the standard deviation of the daily volume for the last 60 days

$\mathcal{T} \leftarrow \mathcal{T} \cup \text{Gathering Episodes}(\text{date}, \mu, \sigma)$

end for

Using $\mathcal{T}$, update $\mathcal{E}$, D by optimizing the loss function J_{TF} in (5) with respect to $\theta^{\mathcal{E}}$ and θ^D

Optimize the loss function J_{PPO} in (9) with respect to $\theta^{\mathcal{L}}$

$\mathcal{T} \leftarrow \emptyset$

end for

Algorithm 2 Gathering Episodes

Require: The number of intervals in a day L

Input: Date, μ , σ

Output: Trajectory buffer of i^{th} day $\mathcal{T}_i$

Initialize $\mathcal{T}_i$

$O \sim \mathcal{N}(2.5 \times 10^{-3} \mu, 6.25 \times 10^{-6} \sigma^2)$

$E_{\text{out}} = \mathcal{E}(E_{\text{in}})$

for $l = 0$ to $L - 1$ **do**

if $l = 0$ **then**

 Construct D_{in}^0 from raw pre-market data

else

$D_{\text{in}}^l \leftarrow D_{\text{in}}^{l-1} \cup \text{Concat}(D_{\text{out}}^{l-1}, h_{T-1}^{l-1})$

end if

Obtain u_{pred}^l and D_{out}^l by procedure depicted in Section 3.3.1

$O^l \leftarrow O \cdot u_{\text{pred}}^l$

Obtain $\tau_l = \{s_t^l, a_t^l, r(a_t^l), V(s_t^l), \pi(a_t^l | s_t^l), O_t^l\}_{t \in \{0, 1, 2, \dots, T-1\}}$ and h_{T-1}^l by procedure depicted in Section 3.3.2

$\mathcal{T}_i \leftarrow \mathcal{T}_i \cup \tau_l$

end for

stock, we account for the differences in total orders for various stocks, which is often encountered in real-world scenarios. Additionally, during testing, we only use the daily ground truth U-shape volume ratios of dates in the training data for our Transformer Encoder to prevent looking ahead into the future. As in previous works of Nevmyvaka et al. (2006), Hendricks and Wilcox (2014), Ning et al. (2021), and Lin and Beling (2020), we make the following assumptions in our experiments.

1. We assume that the actions taken by our model only affect a temporary market, and that the market will recover to the equilibrium level at the next time step.

Table 2

Average volume for each stock from January 1st, 2021 to December 31st, 2021.

Stocks	SE	SH	KC	PH
Volume	17,000K	3800K	3900K	430K

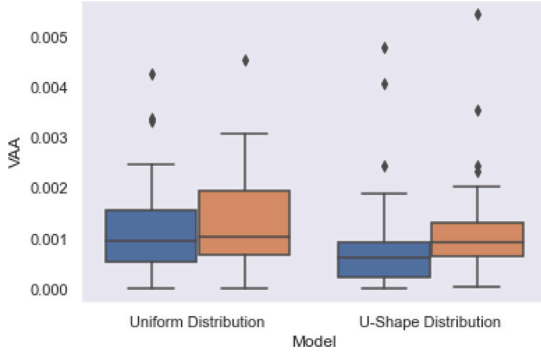


Fig. 5. The VAA's are depicted in dark blue for the best-case scenarios of uniform distribution and U-shape distribution order allocation schemes, while their worst-case counterparts are shown in brown. The uniform distribution and U-shape distribution allocate total orders to be executed per minute in the same manner as PPO and (HU)-PPO, respectively. .

2. Commissions and exchange fees are ignored.
3. Our model's orders are traded immediately without order arrival delays.

We believe that the above assumptions are reasonable as we consider relatively small total orders in comparison to the daily total market volumes of the stocks.

4.1.2. Datasets

Our millisecond trade and limit order book (LOB) tick data is from the Korea Exchange (KRX). We use the daily trade and LOB data of Samsung Electronics (SE), SK Hynix (SH), Kia Corporation (KC) and POSCO Holdings Inc (PH) from January 1st, 2021 to December 31st, 2021. Our choice of the four firms in the KOSPI index is based on their liquidity and trade volume and variety in market capitalization and industry. In order to evaluate the effectiveness of our approach in relation to trading volume, we have opted to select stocks with a range of trade volumes. Specifically, we selected stocks representing different trade volume levels — SE for large, SH and KC for medium, and PH for small volumes. The average daily trading volumes for each stock can be found in Table 2. We divide the data into two sets, using January 1st to September 30th as training data and October 1st to December 31st as test data.

The millisecond raw data is preprocessed by first dividing them into groups of 5 s, and extracting data that represents each 5-second interval. We take the last LOB data of the 5-second interval to construct the MDP state and the 5-second VWAP and total traded volume for calculating the daily VWAP. The statistical U-shape statistics are constructed by taking the year average of the U-shape ratios of the 20-minute intervals in the day. The data utilized in generating D_{in}^0 shown in Section 3.3 are the top 5 bid/ask volume averages of the raw pre-market data (from 08:30:00 to 09:00:00).

4.1.3. Comparison and ablation study

We provide a brief explanation of all methods used for performance comparison during our experiments here. The hyperparameters for each method can be found in Table 3.

- **DQN** is the method proposed in Ning et al. (2021). To apply this model, we begin by dividing a given day into six 1-hour intervals, excluding the final 20 min of market hours to replicate

the experimental setup in Ning et al. (2021). The total order for the day is then equally distributed amongst each hour, and **DQN** further allocates the order for five sub-intervals of 12 min each. The orders are executed equally for each 5-second time step within the 12-minute sub-intervals. To ensure fair comparisons, we evaluate the daily price generated by **DQN** against the daily cumulative VWAP calculated by excluding the final 20 min.

- **OPD** is the method proposed in Fang et al. (2021). To implement this model, we first divide a given day into ten intervals, each spanning 38 min, and then use **OPD** to allocate orders for each interval. Within each interval, the allocated order is equally distributed amongst each minute.
- **PPO** is the method proposed in Lin and Beling (2020). To apply this model, we equally distribute the day total order to each minute in the day, i.e. $O_j = O/380$ for all $j \in \{1, \dots, 380\}$ as in Lin and Beling (2020).
- **(HU)-PPO** utilizes the statistical U-shape to determine O^l , and within each interval, the allocated order is equally divided among every minute, i.e. $O_t^l = O^l/20$ for all $t \in \{0, \dots, 19\}$, where PPO is finally used to distribute orders within the minute.
- **HUL** is our proposed dual-level approach which uses the statistical U-shape to determine O^l .
- **TUL** is our proposed dual-level approach that employs the Transformer model to determine O^l .
- **TUR** uses RNNs to allocate orders within each interval (second level) instead of LSTMs. Other configurations are the same as that of **TUL**.

We evaluate the performance of these methods alongside our proposed dual-level approach based on the VAA metric in (7). As we mentioned in Section 3.3.3, the VAA directly measures the ability of a trading strategy to closely track and approximate the daily VWAP. Consequently, to comprehensively assess ability of each method to consistently track the daily VWAP over an extended period, we analyze both the mean and variation of the VAA metric.

4.2. Experiment results

We test our models, which are trained for 100,000 iterations, by conducting trading simulations in the buying direction. We first verify the claim of the ineffectiveness of short trading horizons when order allocation does not consider the appropriate distribution method for volumes from long trading horizons.

To do this, we examined two distinct scenarios: firstly, we uniformly distributed volumes from a long trading horizon into a short trading horizon and secondly, we employed a statistical U-shape distribution to allocate volumes from a long trading horizon into a short trading horizon. Note that the volumes from a long trading horizon mean daily total orders, while a short trading horizon represents a 1-minute trading horizon. To compare the best and worst cases, we deliberately executed trades for the lowest and highest buy orders, respectively, within the 1-minute trading horizon. Fig. 5 displays the comparison of the VAA's on the test dates. We can observe that the worst case for U-shape distribution outperforms the best case for uniform distribution. Specifically, the VAA's mean of the worst case for U-shape distribution is 10 bps while the VAA's mean of the best case for uniform distribution is 11 bps. This result suggests that in the context of VWAP, the use of deep reinforcement learning does not significantly improve the approximation capability without taking into account the long-term trading horizon. This implies that in order to effectively leverage the benefits of deep RL in the context of VWAPs, it should be implemented on a longer trading horizon such as a single trading day

Secondly, we present Table 4 and Fig. 6 to demonstrate how our dual-level approach significantly improves performance.

Table 4 reports the mean and standard deviation values of VAA's produced by DQN, OPD, PPO, HUL and TUL across the test dates,

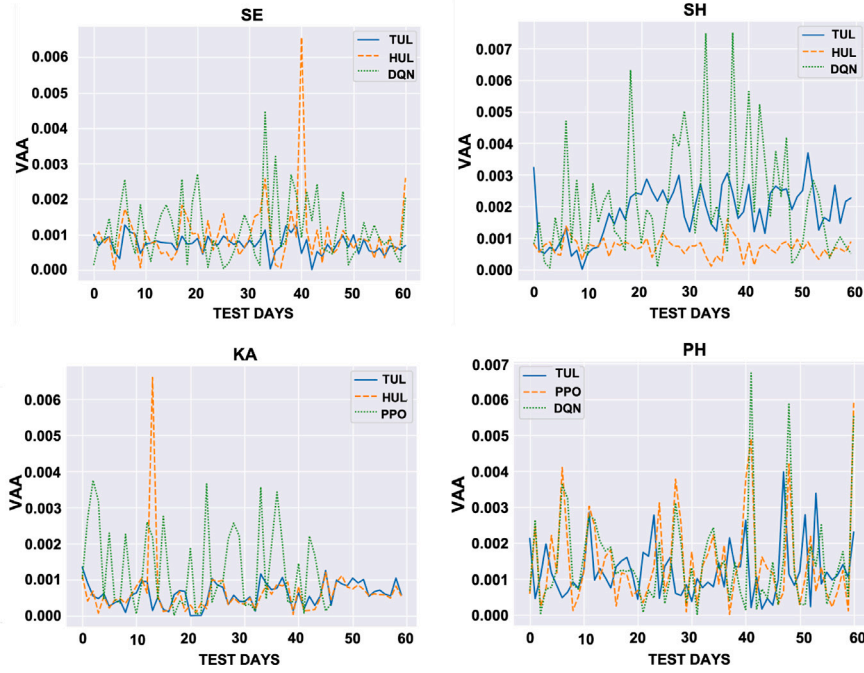


Fig. 6. VAA for the top three models as per the evaluations in Table 4, over a period of 60 test days for selected stocks. The models include TUL, HUL, and the best-performing third model specific to each stock. Notably, TUL demonstrates more consistent VAA with fewer fluctuations compared to other models across all stocks.

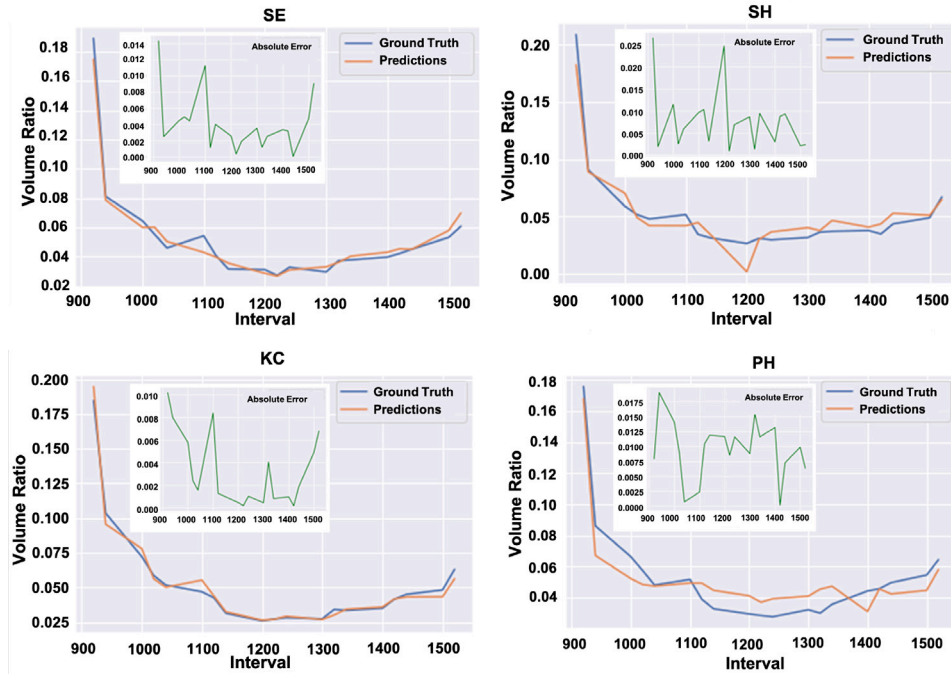


Fig. 7. U-shape averages on test days for ground truths and Transformer-predicted values. The inner plots depict the absolute error values of the predictions for each stock.

and the percentage of them that fall under 10 bps. These methods are used by practitioners, to measure how the traders were performing relative to the VWAP. Our proposed dual-level approach demonstrates significant improvements compared to using short trading horizons and single-level neural networks. Although HUL occasionally yields better accuracy, TUL generally has much less deviation from the daily cumulative VWAP than all other methods. This becomes evident in the less liquid stock PH stock, where HUL lagged behind both DQN and PPO. In this case, DQN and PPO exhibited lower mean and lower standard deviation compared to HUL. Since the daily volume variations

are significant for stocks with small volumes, it is challenging to obtain accurate predictions of the daily volume distribution by only using a statistical U-shape distribution. However, U-shape Transformer in the first level can properly reflect the daily variation, and TUL outperforms both models and demonstrates the lowest mean and standard deviation values. The histograms for VAA distributions given by DQN, OPD, PPO, HUL and TUL on each stock test data are drawn in Fig. 9, which depicts the clustering tendency of accuracies produced by TUL.

Thirdly, we compare the performance of TUL and TUR to examine which neural network architecture is optimal for order allocation in

Table 3

Hyperparameters for DQN, OPD, PPO, HUL and TUL. Note that $\backslash$ indicates a linearly annealing learning rate schedule. Network configurations of DQN and OPD follow Ning et al. (2021) and Fang et al. (2021), respectively.

Hyperparameter	DQN	OPD	PPO	HUL	TUL
Outer iterations	10 000	10 000	10 000	10 000	10 000
Inner iterations	20	10	10	10	10
Batch size	50	10	20	10	10
PPO CLIP ϵ	–	0.2	0.2	0.2	0.2
Discount factor γ	–	1	1	1	1
Number of trajectories per each outer iteration	1000	160	200	152	228
LSTM hidden dimension H	–	–	128	128	129
LSTM model learning rate	–	–	$5e-5 \backslash 1e-5$	$5e-5 \backslash 1e-5$	$5e-5 \backslash 1e-5$
DQN or OPD model learning rate	$1e-4$	$1e-4$	–	–	–
U-shape Encoder input vector dimension N	–	–	–	–	20
U-shape Encoder embedding dimension	–	–	–	–	148
Transformer Encoder & Decoder number of heads	–	–	–	–	4
Transformer Encoder & Decoder number of layers	–	–	–	–	1
Transformer Encoder & Decoder PFFN dimension	–	–	–	–	128
Transformer model learning rate	–	–	–	–	$1e-3 \backslash 2e-4$

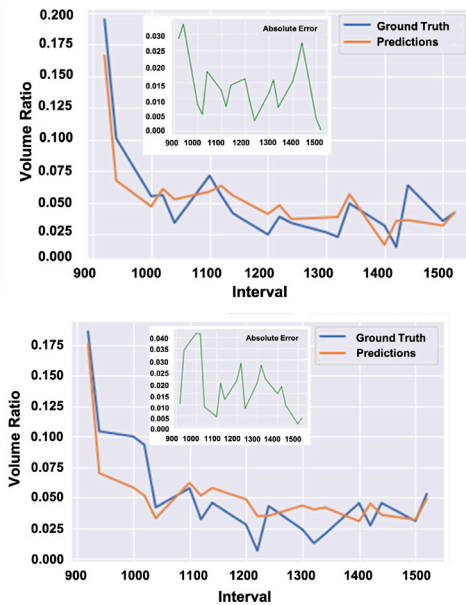


Fig. 8. The ground truth and Transformer-predicted U-shape plots for two selected test dates for PH. The inner plots depict the absolute error values of the predictions.

the second level. For SE, which is the stock with the highest trading volume, TUR slightly underperforms HUL and TUL and outperforms all other models. However, it is evident that a significant decline in performance occurs for all other stocks with lower trading volumes. This performance degradation can be attributed to the limited capability of the simpler RNN model when compared with the LSTM model to capture volume fluctuations in stocks that exhibit relatively lower liquidity.

Lastly, we demonstrate the effectiveness of the proposed U-shape Transformer using Figs. 7 and 8. Fig. 7 shows the plots of the ground truth U-shape and Transformer-predicted U-shape values, where the U-shape ratio values for each 20-minute interval were averaged across all test dates. We observe that our U-shape Transformer is able to follow the U-shape pattern for the test dates on average for all stocks, which is in line with the comparable (and sometimes better) performance of TUL against HUL. This is further supported by the fact that HUL only utilizes the average U-shape ratio values of the past, while TUL

approximates the U-shape distribution for the test dates accurately and uses predicted values for order allocation. Thus, TUL becomes more adaptive in tracking the daily cumulative VWAP as it incorporates predictions of fluctuations that may appear in the U-shape distribution in the future.

We also validate the performance improvements of TUL compared to other models, particularly HUL, by examining TUL's ability to capture daily fluctuations in the U-shape. Fig. 8 displays the ground truth U-shape and Transformer-predicted distributions for two sampled test dates for the stock PH. We deliberately choose PH since the performance gains of TUL when compared to HUL were the largest for this particular stock, which has daily volume distributions that deviates the most from the average due to relatively low liquidity. While the U-shape Transformer may occasionally make erroneous predictions, it quickly corrects itself and adapts to changes in the daily volume distribution, attempting to accurately follow ratio changes throughout the day.

5. Discussion and conclusion

In this study, we propose a dual-level approach to address the problem of tracking the VWAP with accuracy and consistency. Initially, as observed in Fig. 5, we identified that the naive uniform distribution approach falls short in accurately tracking the daily VWAP. The uniqueness of our model is in how we improve upon this issue by taking into account the well-known U-shaped intraday trading pattern. In the first stage, we allocate the number of orders to execute in each interval based on the U-shape pattern. We consider two methods, statistical U-shape and U-shape Transformer, for implementing the first stage. The naive approach of using the statistical U-shape distribution improves performance. However, our experiments demonstrate that the U-shape Transformer performs even better. We believe that the superior performance of the U-shape Transformer is due to its ability to adapt to unexpected variations, a capability that the statistical U-shape method lacks, as evidenced by Figs. 7 and 8. After the total daily orders are allocated to each interval in the first stage, the LSTM model is used in the second stage to determine how the orders in each interval should be executed. Our simulation results show that the dual-level approach consistently and accurately tracks the daily cumulative VWAP.

While our method demonstrates superior performance through experiments, a comprehensive mathematical understanding of its underlying principles including uncertainty quantification remains to be explored. The limitations identified in our paper suggest potential

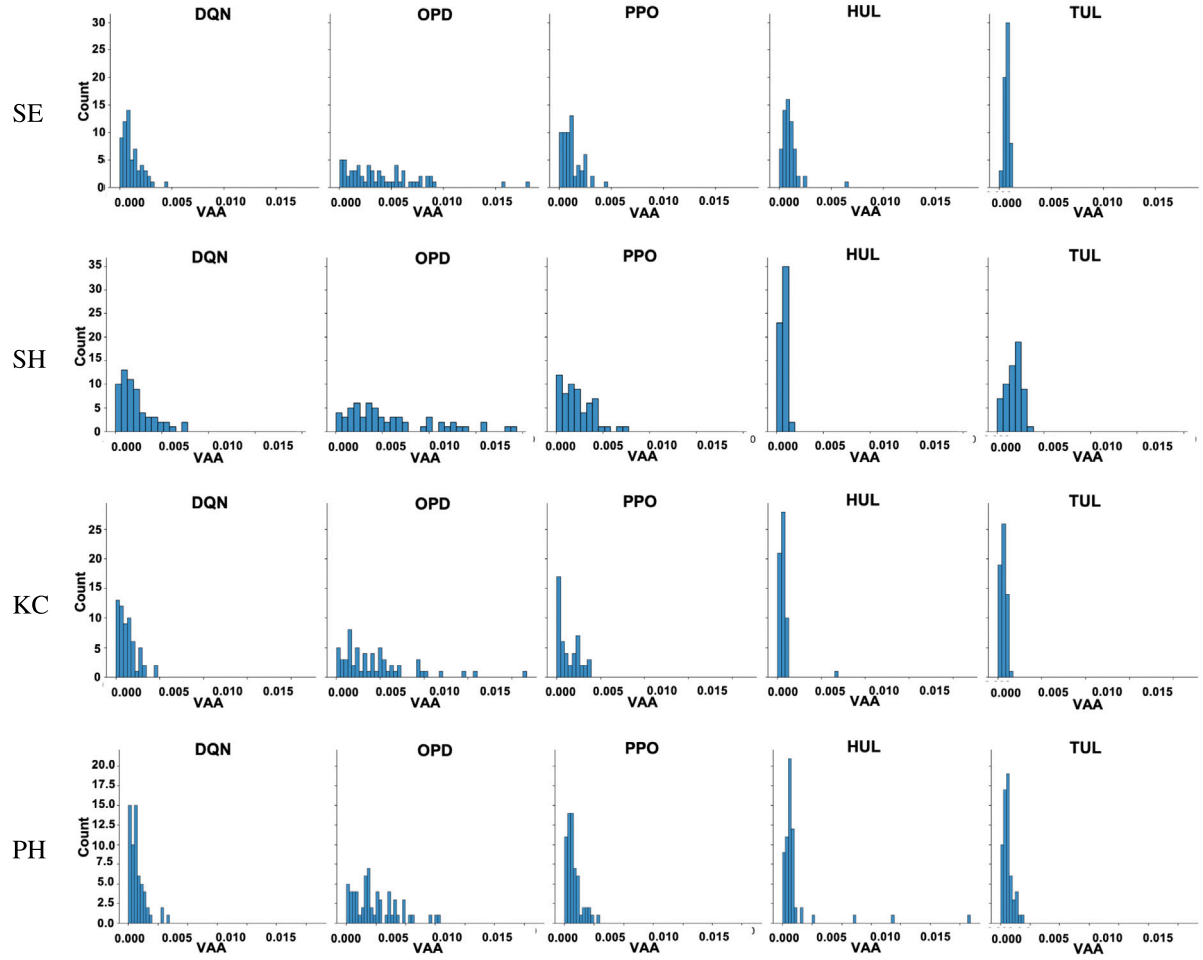


Fig. 9. The histograms of the VAA distributions for each model and stock on the test dates. Both TUL and HUL, employing the U-shape property, exhibit consistent stability in tracking the VWAP for SE, SH, and KC stocks. Moreover, TUL demonstrates robust tracking for the PH stock, which is characterized by significant daily volume fluctuations.

Table 4

Table 4 summarizes the mean, standard deviation, and percentage of VAAs within the range [0, 10 bps]. The Mean and Standard Deviation are represented in units of basis points (bps), and the % in 10 bps are represented in units of percentages. For OPD, we excluded VAAs above the 80th percentile for fair comparisons since our training dataset was smaller than that used in Fang et al. (2021). For TUR, we excluded extreme outliers for SH, KA, and PH.

Stocks	Metric	OPD	DQN	PPO	HUL	TUL	TUR
SE	Mean	26.27	11.53	12.00	8.77	7.13	9.43
	Standard Deviation	18.50	8.84	8.88	7.17	2.44	9.56
	% in 10 bps	25.00	57.38	49.18	65.57	88.52	65.57
SH	Mean	38.55	21.62	22.57	7.13	7.65	48.56
	Standard Deviation	24.90	17.71	16.92	2.81	3.63	33.51
	% in 10 bps	10.42	30.00	25.00	91.67	88.33	5.45
KA	Mean	27.79	13.16	12.47	6.59	6.03	149.32
	Standard Deviation	18.76	10.25	11.55	8.36	3.30	111.64
	% in 10 bps	18.75	46.67	56.67	90.00	86.67	9.43
PH	Mean	32.44	14.84	14.64	22.52	12.27	377.28
	Standard Deviation	20.89	13.47	12.63	47.11	7.90	267.23
	% in 10 bps	18.75	40.98	40.98	32.79	44.26	1.85

directions for future research. Additionally, we note that our proposed method entails the following ethical implications. Firstly, our approach provides investors with new options for dealing with market volatility, helping to avoid market impact and the risk of front-runners exploiting their strategies. Secondly, improved ability of our execution model to track the VWAP contributes to mitigating market impact, which can potentially enhance market transparency and fairness.

CRediT authorship contribution statement

Soohan Kim: Writing – original draft, Methodology, Software.
Jimyeong Kim: Writing – original draft, Data curation, Conceptualization, Visualization.
Hong Kee Sul: Writing – review & editing, Investigation.
Youngjoon Hong: Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Youngjoon Hong reports was provided by Sungkyunkwan University.

Data availability

Data will be made available on request.

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